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MILITARY SPECIFICATION

TORPEDO TUBE, SURFACE VESSEL, MARK 32 MODS 5 AND 7

(For Torpedoes Mark 44 Mod 1 and Mark 46 Mod 1)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 This specification establishes the requirements for the procurement of Torpedo Tube Assembly Mk 32 Mod 5 (NAVSEA LD 162165) and Torpedo Tube Assembly Mk 32 Mod 7 (NAVSEA LD 162166) for use on surface vessels.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation - Packaging, Methods of
MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and *Systems*, Requirements for
MIL-C-16173	Corrosion Preventative Compound, Solvent Cutback, Cold-Application

FSC 1045

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

Naval Sea Systems Command (Code Ident 10001)

NAVSEA LD 162135	Torpedo Tube, Surface Vessel, Mk 32 Mod 5 (Assembly)
NAVSEA LD 162155	Torpedo Tube, Surface Vessel, Mk 32 Mod 7 (Assembly)
NAVSEA LD 162165	Torpedo Tube, Surface Vessel, Mk 32 Mod 5 (Assembly), Index to List of Drawings
NAVSEA LD 162166	Torpedo Tube, Surface Vessel, Mk 32 Mod 7 (Assembly), Index to List of Drawings
NAVSEA Dwg 694718	Torpedo Tube, Surface Vessel, Mk 32 Mod 5 (Assembly)
NAVSEA Dwg 694720	Torpedo Tube, Surface Vessel, Mk 32 Mod 7 (Assembly)
NAVSEA Dwg 694627	Torpedo Tube, Surface Vessel, Mk 32 Mods 5 and 6, Electrical Schematic
NAVSEA Dwg 694631	Torpedo Tube, Surface Vessel, Mk 32 Mods 7 and 8, Electrical Schematic
NAVSEA Dwg 694740	Torpedo Tube, Surface Vessel, Mk 32 Mods 5 and 6, External Wiring Diagram
NAVSEA Dwg 694741	Torpedo Tube, Surface Vessel, Mk 32 Mods 7 and 8, External Wiring Diagram

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NAVSEA Dwg 1623922	Pressure Gage
NAVSEA Dwg 1984119	Dummy Torpedo Mk 44 Mod 1
NAVSEA Dwg 2320787	"GO" Gage, Barrel (Assembly) Torpedo Tube, surface Vessel, Mk 32 Mods O thru 9
NAVSEA Dwg 2652289	Dummy Torpedo Mk 46 Mod 1
NAVSEA Dwg 3245120	Pressure Fitting Adapter

(copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

National Bureau of Standards Publications

Handbook H28	Screw Thread Standards for the Federal Services

(Application for copies should be addressed,to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. Torpedo Tube Assemblies Mk 32 Mods 5 and 7 are pneumatic launching devices capable of receiving, stowing, and launching Torpedoes Mk 44 Mod 1 and Mk 46 Mod 1. The tube assemblies consist of three parallel barrels in triangular configuration, with three Breech Mechanisms Mk 1 Mod 1 and torpedo securing mechanisms, each an integral part of the barrels which are nested together on a common Training Gear Mk 11 Mod 1. Each barrel is capable of being selected and fired electrically from a remote location and of being manually fired at the tube in an emergency condition. During manual firing, a latch plate is swung open *to* provide automatic switching from normal to emergency torpedo gyro squib fire power for Torpedo Mk 46 Mod 1 and expose a manual firing button to actuate the solenoid valve. The tube is equipped with two electrical junction boxes, one on each of the two lower barrels. One box receives electrical presetting functions from the fire control system and transmits them via electrical

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cabling to each barrel-housed torpedo. The other box receives and distributes the barrel ready, firing, and heating circuits. A charging line stowage tube is mounted on the uppermost barrel to accommodate the flexible charging hose. The only difference between the Mods 5 and 7 is in the electrical circuits: the Mod 5 is compatible with Fire Control Systems Mk 102, 105, and 111; the Mod 7 is compatible with Fire Control System Mk 114.

3.2 General requirements

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order(see 6,2), a preproduction sample of the torpedo tube assembly is required and shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in 4.3.1 for the purpose of determining that, prior to starting production the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2.2 Materials. Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of Torpedo Tubes Mk 32 Mods 5 and 7 shall conform to the detailed requirements given on the drawings and specifications listed in NAVSEA LD 162135 and LD 162155, respectively.

3.2.3 Finishes,and coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions shall be in accordance with the applicable drawings, specifications, and publications listed in NAVSEA LD 162135 and LD 162155, respectively.

3.2.4 Moisture, fungus, and corrosion. The tubes shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on drawings and specifications in NAVSEA LD 162135 and LD 162155, respectively.

3.2.5 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in NAVSEA LD 162135 and * LD 162155, respectively.

3.2.6 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or * fit for threads shall be as specified on the applicable drawings.

*

3.2.7 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.2.8 Safety precautions. All standard safety precautions for electrical apparatus and high pressure air shall be strictly observed, No test shall

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be permitted to continue if in the opinion of the test engineer it is deemed unsafe. While conducting any of the firings, personnel shall stand clear of the breechward access opening and breech mechanism. Charging lines should be secured with safety chains when high pressure air charging operations are in progress.

3.3 Definitions. For the purpose of this specification the following definitions are applicable:

3.3.1 Torpedo tube assembly. The launching device in its assembled state.

3.3.2 Nonoperating condition. The condition of the tube when assembled, prepared, issued, transported, and handled.

3.3.3 Loaded condition. That condition when a torpedo or dummy is installed in the tube barrel, secured in place by the securing mechanism, and the breech mechanism and muzzle covers are installed.

3.3.4 Latched position of the breakaway plug puller mechanism. The breakaway plug puller mechanism is latched when the latch is in the breechward position.

3.4 Performance requirements and product characteristics. Torpedo Tubes MK 32 Mod 5 and Mk 32 Mod 7 shall meet the following performance requirements and product characteristics:

3.4.1 Hydrostatic pressure (breech mechanisms). The breech mechanisms shall withstand a hydrostatic pressure of 4500 plus 20 minus 0 psig without failure of any part.

* 3.4.2 Pneumatic system pressure drop. The pneumatic system of each barrel of the torpedo tube, minus the breech mechanism, shall be attached to a 2000 psi air source through a 0.024 inch diameter orifice. With the securing mechanism removed and the manifold nipple blanked off, the pressure in the pneumatic system of each barrel shall stabilize at 1350 psig minimum. With the securing mechanism installed, the pressure shall stabilize at 750 psig minimum. At the completion of this test there shall be no deformation or failure of any part of the pneumatic system.

3.4.3 Breech mechanism air loss. After being initially charged to a pressure of 2000 plus or minus 50 psig, allowed to cool for 2 hours, and then bled down to 1600 plus or minus 10 psig, none of the three breech mechanisms shall lose more than 5 percent air pressure, due to leakage, over a 24-hour period.

* 3.4.4 Bore size of barrel. Each barrel shall pass a “GO” gage (NAVSEA Dwg 2320787) with an axial force not to exceed 150 pounds.

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3.4.5 Electrical requirements

3.4.5.1 Dielectric strength. Electrical circuitry of the tube shall withstand the application of a potential of 1000 volts ac between mutually insulated terminals and ground for one minute minimum without breakdown.

* 3.4.5.2 Insulation resistance. Immediately following the dielectric strength test, the insulation resistance between hipotted points in the barrel circuits shall be 3 megohms minimum when a potential of 500 volts dc is applied. The insulation resistance between the hipotted points of the training gear heater circuit shall be 0.5 megohm minimum when a potential of 500 volts dc is applied.

3.4.5.3 Circuitry. Electrical circuitry shall be in accordance with NAVSEA Dwg 694627 for Torpedo Tube Mk 32 Mod 5 and NAVSEA Dwg 694631 for Torpedo Tube Mk 32 Mod 7, as applicable.

* 3.4.5.4 Training gear heater resistance test or current drain. The torpedo tube assembly training gear heater element resistance shall be 6.9 ohms plus 0.7 ohm or minus 0.4 ohm, or the heater element current drain at 115 volts plus or minus 0.5 volt ac input shall be 16.7 amps plus 1.1 amps or minus 1.6 amps.

* 3.4.5.5 Barrel heater resistance. The total electrical resistance of the heater circuit for each barrel shall be 11.4 plus or minus 0.8 ohms.

* 3.4.5.6 Heater circuits operating current. When the electrical heating circuits of the torpedo tube are operating, the total maximum current drain shall be 50.5 amperes with an input of 115 plus or minus 0.5 volts ac.

3.4.6 Squib fire switches. The switch plunger shall move out forcefully when the latch plate is released.

3.4.7 Interlock switch. When the training gear is rotated outboard through 180 degrees from the zero degree relative position, the training gear interlock switch shall be closed from 35 plus or minus 2.5 degrees to 145 plus or minus 2.5 degrees.

3.4.8 Training gear crank test. When the base of the torpedo tube is level, the force required to initiate movement of the training gear, applied tangentially to the arc described by the training gear crank handle, shall be not greater than 30 pounds.

3.4.9 Retractable stop. The retractable stop, when in the operating position, shall stop tube rotation at 45 degrees plus or minus 2.5 degrees from the zero position.

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3.4.10 Limit stops. The limit stops shall prevent and arc of train in excess of 190 degrees.

3.4.11 Muzzle cover seal test. The axial force required to remove the muzzle cover shall be 30 plus or minus 10 pounds.

3.4.12 Ready light switch. When the switch actuator plate on each control box is placed in the ON position, the ready light switch contacts shall close.

* 3.4.13 Pressure switch operation. The pressure switch contacts in each barrel shall close at 1275 psig plus or minus 75 psig with increasing pressure and remain closed at pressures above this value. The pressure switch contacts shall open at 1275 plus or minus 75 psig on falling pressure and remain open at pressures below this value.

3.4.14 Breakaway plug puller mechanism

3.4.14.1 The lever of the breakaway plug puller mechanism shall be capable of being latched and unlatched by hand.

3.4.14.2 The breakaway plug puller mechanism shall permit dry firings when in the latched position and shall remain in the latched position at the completion of these firings.

* 3.4.14.3 The breakaway plug, during torpedo firings, shall pull away from the torpedo receptacle in Torpedoes Mk 44 Mod 1 and Mk 46 Mod 1 without damage and shall not affect torpedo exit velocities when the tube is fired at pressures of 1200 through 1500 psig.

3.4.15 Control valve operating lever. Each barrel of the tube shall not be capable of being fired, either electrically or manually, until the control valve operating lever is placed in the READY condition.

3.4.16 Solenoid voltage. The solenoid firing valve for each barrel shall operate throughout an input voltage range of 102 to 124 volts ac, 60 Hz, single phase.

* 3.4.17 Manual firing. The axial force that must be applied on the manual firing button to initiate the firing cycle at pressures of 1200 through 2000 psig shall not exceed 25 pounds.

* 3.4.18 Dry firing test. Each barrel of the torpedo tube assembly shall be capable of being fired, without a torpedo, by mechanical means at any flask pressure from 1200 to 2000 psig and by electrical means when the solenoid valves are energized for 400 milliseconds maximum at any flask pressure of 1350 through 2000 psig without damage, malfunction, or hindrance to subsequent firings with a torpedo.

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3.4.19 During dry firings of the tube and firings of dummy Torpedces Mk 44 Mod 1 and Mk 46 Mod 1 at pressures of 1200 through 1500 psig, the plug puller mechanism shall move automatically from the unlatched to the latched position. Upon completion of firings, the lever shall be capable of being unlatched by hand.

3.4.20 Breech mechanism interchangeability. Each breech mechanism assembly shall be capable of operating within every barrel of the tube assembly.

* 3.4.21 Ready-light effect on electrical firing capability. Each barrel of the torpedo tube assembly shall be capable of being electrically fired at 1350 psig minimum under conditions that will close all three switches in the ready light circuit (firing interlock, pressure, and ready light).

* 3.4.22 Torpedo exit velocities. The exit velocities of dummy Torpedoes Mk 44 Mod 1 (433 plus or minus 7 pounds) and Mk 46 Mod 1 (508 plus or minus 8 pounds) for each barrel of the torpedo tube assembly shall be in accordance with Schedule II as specified in 4.4.5.23.

3.4.23 Electrical continuity

3.4.23.1 Electrical continuity of Torpedo Tube Assembly Mk 32 Mod 5 shall be in accordance with NAVSEA Dwg 694627.

3.4.23.2 Electrical continuity of Torpedo Tube Assembly Mk 32 Mod 7 shall be in accordance with NAVSEA Dwg 694631.

* 3.4.24 Lanyard retainers. The spring-loaded retainers shall be capable of being depressed by hand and shall be capable of accepting and securing a lanyard wire of 0.069 minimum diameter.

3.5 Environmental requirements

3.5.1 Vibration

3.5.1.1 Vibration, nonoperating. The torpedo tube assembly, less packing or packaging, shall be capable of meeting all quality conformance tests after being subjected to vibration tests of MIL-STD-167 for Type I vibrations in the three mutually perpendicular axes.

3.5.1.2 Vibration, standby condition. The torpedo tube assembly, loaded with three dummy torpedoes and containing breech assemblies pressurized to 1500 psig, shall be capable of meeting all quality conformance tests during and after being subjected to the vibration tests of MIL-STD-167 for Type I vibrations in the three mutually perpendicular axes.

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3.5.2 Shock

3.5.2.1 Shock, nonoperating. The torpedo tube assembly, less packing or packaging, shall be capable of meeting all quality conformance tests after being subjected to the shock impulses of the magnitude and number specified in MIL-S-901 for medium weight equipment.

3.5.2.2 Shock, loaded condition. The torpedo tube, loaded with three dummy torpedoes, shall be capable of meeting all quality conformance tests after being subjected to the shock impulses of the magnitude and number specified in MIL-S-901 for medium weight equipment.

3.5.3 Temperature and humidity

* 3.5.3.1 Low temperature operation, loaded condition. The torpedo tube, less packaging and packing and in the loaded condition, shall be capable of withstanding the low temperature conditions specified in 4.5.3.1. While at the low test temperature; the training gear heater shall maintain the training gear operable and free from obstruction due to ice and the barrel heaters shall maintain the temperature on the outside surface of the torpedo at not less than 0.5°C (33°F). At the conclusion of the test and while still at the test temperature, the torpedo tube shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

* 3.5.3.2 High temperature operation. The torpedo tube, with fully pressurized breech assemblies, shall be capable of withstanding the high temperature and humidity conditions specified in 4.5.3.2. While at high temperature and humidity conditions, the torpedo tube shall meet all the performance requirements and product characteristics of this specification. The barrel heater relay shall open the heater circuit during high temperature conditions and close it during low temperature portions of all cycles. At the conclusion of this test, the torpedo tube shall show no corrosion or damage.

* 3.5.4 Marine atmosphere environment. The torpedo tube shall be capable of withstanding the marine atmosphere or alternative test as specified in 4.5.4 and thereafter shall show no evidence of corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.6 Cleanup. Prior to and after final assembly, all parts and components shall be thoroughly cleaned of all loose, spattered, or excess solder, metal chips, and other foreign material. All burrs and sharp edges as well as flashing shall be removed.

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3.7 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils , marking of parts and assemblies, plating, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved pre-production sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The examination and testing of Torpedo Tubes Mk 32 Mods 5 and 7 shall be classified as follows:

a. Preproduction inspection. The inspection of the preproduction sample submitted in accordance with 4.3.1 (See 4.5).

b. Quality Conformance inspection. Tests and examinations performed on each torpedo tube assembly to be submitted for acceptance under contract (See 4.4).

c. Periodic production inspection. The inspection of periodic production samples submitted in accordance with 4.3.3 (See 4.5).

d. Post-delivery inspection. Unless otherwise specified in the contract or order (See 6.2), the Government reserves the right to conduct any or all of the tests and examinations of this specification after installation of the torpedo tube aboard a Naval vessel. Any defects due to the contractor's failure to comply with drawings or specifications for this equipment, discovered during or after these tests or examinations, shall be corrected by the contractor at no expense to the Government , provided the contractor is notified of such defects within the guarentee period of the contract.

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4.3 Preproductions and periodic prodution samples

* 4.3.1 Preproduction sample. A preproduction sample of one torpedo tube assembly, manufactured in accordance with 3.2.1 and preserved, packaged and-packed as specified in the contract or order, shall be subjected to the preproduction tests detailed in 4.5, at an activity designated by the procuring activity (See 6.2). Further production of the torpedo tube by the contractor prior to approval of the pre-production sample shall be at the contractor's own risk. Approved production samples shall be included as a part of the quantity specified for delivery in the contract or order.

* 4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

* 4.3.3 Periodic production sample. From each succeeding lot of 50 or less, a periodic production sample of one torpedo tube assembly shall be selected at random and submitted tp an activity designated by the procuring activity (See 6.2) for periodic production inspection. The sample shall be examined for compliance with the requirements of the contract, specification and drawings and tested in accordance with 4.5. Failure of the sample to comply with any requirement shall result in the rejection of the lot as determined by the procuring activity (See 6.2).

* 4.4 Quality conformance inspection. Each torpedo tube Mk 32 Mod 5 and Mk 32 Mod 7 to be delivered under contract shall be subjected to the tests and examinations of this section.

4.4.1 Test conditions. unless otherwise specified, the torpedo tubes shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Test sequence. The sequence of testing procedures presented herein is suggested only. The tests may be performed in any order that will permit the most practical and economical use of manpower and equipment.

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4.4.1.2 Temperature. Room ambient 65 degrees Fahrenheit (F) to 95 degrees F (18 degrees centigrade (C) to 35 degrees C).

4.4.1.3 Altitude. Normal ground atmospheric pressure.

4.4.1.4 Vibration: None.

4.4.1.5 Humidity: Room ambient to 95 percent relative maximum.

* 4.4.2 Test results. When specified in the contract or order (See 6.2), test results shall be recorded, or verification noted, as required, opposite corresponding paragraph numbers on the Quality Conformance Inspection Data Sheet Appendix) for each torpedo tube assembly inspected (See 5.4).

4.4.3 Acceptance criteria. Failure of Torpedo Tube Mk 32 Mod 5 or Mk 32 Mod 7 to pass any examination or test of this section shall be cause for rejection.

4.4.4 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests. To perform the tests, the following items of test equipment, or equivalent", are required:

4.4.4.1 A means for producing a hydrostatic pressure of 4500 psig.

4.4.4.2 A water pressure gage capable of reading 4500 psig with an accuracy of plus or minus 1/4 percent.

* 4.4.4.3 A 0-3000 psig high pressure air source.

4.4.4.4 An air line filter, *to* remove water, oil, and dirt, which shall not pass particles larger than 50 microns.

* 4.4.4.5 An air pressure gage of O to at least 3000 psig with an accuracy of plus or minus 1./4 percent at 2500 psig.

* 4.4.4.6 A "GO" gage for the barrel in accordance with NAVSEA Dwg 2320787.

4.4.4.7 Standard high potential test equipment capable of an output of 1000 volts, 60 to 100 Hz.

4.4.4.8 Megohm bridge, 500 volts dc, with a range of 0.1 to 1,000,000 megohms.

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* 4.4.4.9 An ohmmeter capable of measuring 11.4 and 6.9 ohms with an accuracy of plus or minus 5 percent of the reading.

* 4.4.4.10 A power supply capable of supplying at least 50.5 amperes at 115 plus or minus 0.5 volts ac, 60 Hz, single phase; the voltage must be capable of being varied from at least 102 to 124 volts ac.

* 4.4.4.11 An ac ammeter capable of measuring currents of 16.7 and 50.5 amps with an accuracy of at least plus or minus 0.375 ampere.

* 4.4.4.12 An ac voltmeter capable of measuring 102.5, 115, and 123.5 volts ac with an accuracy of at least plus or minus 1.125 volts,

* 4.4.4.13 Jumper wires of sufficient length to facilitate performance of the training gear heater resistance test.

* 4.4.4.14 A push scale with a memory pointer and a capacity of at least 30 pounds, with an accuracy of plus or minus 10 percent of the reading.

4.4.4.15 A means for measuring the angle of rotation of the training gear to the nearest 1/2 degree.

* 4.4.4.16 A pull scale with a memory pointer and a capacity of at least 155 pounds, with a maximum error of plus or minus 10 percent of the reading.

* 4.4.4.17 A test circuit capable of supplying 102 to 124 volts, 60 Hz ac from the power supply of 4.4.4.10 to the solenoid for a period of 400 milliseconds maximum. Current capacity shall be at least 2 amperes.

* 4.4.4.18 Dummy Torpedo Mk 44 Mod 1 (NAVSEA Dwg 1984119) (weight: 433 Plus or minus 7 pounds).

* 4.4.4.19 Dummy Torpedo Mk 46 Mod 1 (NAVSEA Dwg 2652289) (weight: 508 plus or minus 8 pounds).

4.4.4.20 Launching ramp or track from which dummy torpedoes can be launched. The ramp or track shall include a means for decelerating and stopping the dummy torpedo on the track without damage.

4.4.4.21 Two infrared photoelectric control units suitable for use with electronic counter of 4.4.4.22.

* 4.4.4.22 An electronic counter capable of recording up to 10,000 milliseconds in one millisecond intervals with instant reset capabilities.

4.4.4.23 A continuity indicator.

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* 4.4.4.24 A $C0_2$ bottle of sufficient volume to cool the thermoswitches for the performance of the dielectric strength test, the barrel heater resistance test, and the heater circuits operating-current test.

* 4.4.4.25 A 125 plus or minus 2 vdc power supply with a 2.7 plus or minus 0.1 k ohm series resistor.

* 4.4.4.26 Pressure fitting adapter (NAVSEA Dwg 3245120).

* 4.4.4.27 Nitrogen bottle of sufficient volume to purge the breech mechanism air flasks.

* 4.4.4.28 A 0.024 plus or minus 0.0005 inch diameter orifice for the high pressure air source of $4.4.4.3_0$

4.4.5 Test procedures. The following tests shall be performed:

WARNING

When working with high pressure air or electrical apparatus, observe all safety precautions (see 3.2.8).

* 4.4.5.1 Hydrostatic Pressure test (breech mechanism). With the breech mechanisms removed from the tube assembly, remove Pressure Gage 1623922 and insert a standard l/8-pipe plug in the void. The breech mechanisms shall be filled with water and pressurized to 4500 plus 20 minus O psig. This pressure shall be maintained for 10 minutes minimum without failure of any part. At the completion of this test reduce the pressure to zero, drain the breech mechanism and purge with dry nitrogen. Reinstall Pressure Gage 1623922.

* 4.4.5.2 Pneumatic system pressure drop test. This test shall be performed on each barrel of the torpedo tube assembly. Remove the breech mechanism and securing mechanism from the barrel. Secure the nipple attached to the manifold and piping assembly to a shutoff valve or other suitable blanking device. Adjust the air source to 2000 plus O minus 20 psig and attach it through a 0.024 plus or minus 0.0005 inch diameter orifice to the barrel pneumatic system using a Pressure Fitting Adapter (NAVSEA Dwg 3245120). Actuate the firing solenoid valve and allow the pneumatic system line pressure to stabilize. The pressure as measured on the barrel side of the orifice shall stabilize at 1350 psig minimum. Repeat this test with the securing mechanism installed in the barrel. The pressure in the pneumatic system shall then stabilize at 750 psig minimum. At the conclusion of this test there shall be no deformation or failure of any part of the pneumatic system.

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* 4.4.5.3 Breech mechanism air loss test. With the breech mechanisms removed, gradually charge each (using the test equipment of 4.4.4.3, 4.4.4.4, and 4.4.4.5) at a rate not exceeding 250 psig per minute to a pressure of 2000 psig pneumatic and permit them to cool for a period of 2 hours. The breech mechanisms shall then be bled to a pressure of 1600 plus or minus 10 psig. After a 24-hour period, observe the pressure reading, using the gage of 4.4.4.5. The pressure loss in each breech mechanism shall not exceed 5 percent Compensation may be made for change in pressure due to temperature.

* 4.4.5.4 Bore size of barrel test. The "GO" bore gage (4.4.4,6) shall pass through each barrel with an axial force not to exceed l5O pounds.

* 4.4.5.5 Dielectric strength tests. For the purpose of this test the barrel and training gear heater switches shall be placed in the ON position and the barrel heater thermoswitches shall be cooled with C02 to close the contacts. Temporarily connect the training gear, and, on the Mod 7 only, the junction box on Barrel Number 2 to terminal 14 of TB 102 (in the junction box on Barrel Number 3). The tube assembly shall withstand a potential of 1000 volts, 60 Hz ac (4.4.4.7) applied between terminal 14 of TB102 and all input terminals as shown on the external wiring diagram, NAVSEA Dwg 694740 (Mod 5) or 694741 (Mod 7), for one minute without evidence of breakdown. The input terminals may be connected together for this test.

* 4.4.5.6 Insulation resistance test. For the purpose of this test the tube assembly shall be in the same condition as required for the dielectric strength test (4.4.5.5) with the exception that the training gear heater switch shall be in the OFF position. Immediately after the hipot test, the insulation resistance between the hipotted points shall be 3 megohms minimum at 500 vdc and the insulation resistance between terminals 1 and 2 of TB102 and terminal 14 of TB102 shall be 0.5 megohm minimum at 500 vdc.

* *4.4.5.7* Ground circuit continuity. Continuity shall exist between terminal 14 of TB102, pin G of test receptacle J2 (in the junction box on Barrel Number 3), and any portion of the junction box as well as anY of the three control boxes and overheat sensor assemblies. On the Mod 5 only continuity shall also be indicated between terminal 14 of TB102 and any portion of the junction box on Barrel Number 2.

* *4.4.5.8* Training gear heater resistance test. Before the tests below are conducted, the thermoswitch shall be jumpered and the training gear heater shall be energized with 115 volts, 60 Hz ac applied at terminals 1 and 2 of TB103 in the junction box for a period of one hour.

* 4.4.5.8.1 With the training gear heater switch in the OFF position, the resistance between pins 1 and 2 of TB102 in the junction box shall be 6.9 ohms plus 0.7 ohms or minus *0.4 ohm.*

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* 4.4.5.8.2 Alternative test. As an alternative test to 4.4.5.8.1, heater current drain may be measured. With the input voltage to the heater element adjusted to 115 plus or minus 0.5 volts, 60 Hz ac, an ac ammeter connected in the circuit, and the training gear heater switch placed in the ON position, the input current shall be 16.7 amps plus 1.1 amps or minus 1.5 amps.

* 4.4.5.9 Barrel heater resistance test. With the barrel heater switches in the OFF position, the two barrel heater thermoswitches in Barrel Number 1 shall be cooled with C02 to close the contacts. The resistance between terminals 3 and 4 of TB101 in the barrel control box shall be 11.4 plus or minus 0.8 ohms. Repeat this test on Barrels Number 2 and 3.

* 4.4.5.10 Heater circuits operating current. For the purpose of this test the 125 volts dc power supply of 4.4.4.25 shall be applied to terminals 5 and 11 of TB2 (Mod 5) or terminals 10 and 13 of TB102 (Mod 7). The training gear and barrel heater switches in the junction box shall be set to the ON position and the thermoswitches in each barrel and in the training gear shall be cooled with C02 to close the contacts. With 115 plus or minus 0.5 volts ac applied to terminals 1 and 2 of TB103 and an ammeter installed in the input circuit, the current in the circuit shall not exceed 50.5 amperes.

4.4.5.11 Squib fire switch. With the control box latch plates in the locked position, continuity shall exist across terminals 8 and 9 of TB101 and no continuity shall exist across terminals 7 and 8 and terminals 7 and 9 of TB101 in the control box of each barrel, Release the latch on the control box, observing that the switch plunger moves out forcefully, and verify that continuity exists across terminals 7 and 8 of TB101 in each barrel and that no continuity exists across terminals 8 and 9, and 7 and 9 of TB101.

4.4.5.12 Interlock switch. The torpedo tube assembly shall be trained through an arc of 180 degrees from the zero degree relative position. Continuity of the switch circuit shall be measured throughout this operation and the angles of open and closed circuit recorded. The switch shall be open at angles of O degrees to 35 degrees plus or minus 2.5 degrees and 145 plus or minus 2.5 degrees to 180 degrees; it shall close at angles of 35 plus or minus 2.5 degrees through 145 plus or minus 2.5 degrees.

4.4.5.13 Training crank test. With the base of the torpedo tube assembly level, a pull scale shall be attached to the training crank handle and pulled with a steady motion tangential to the arc of crank travel. The force required to initiate movement of the training gear shall not exceed 30 pounds

4.4.5.14 Retractable stop. With retractable stop in the operating position, attempt to rotate the torpedo tube assembly through an arc of 180 degrees. The rotation shall be stopped by the retractable stop at 45 degrees plus or minus 2.5 degrees from the zero position.

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4.4.5.15 Limit stops. With the retractable stop in the retracted position rotate the torpedo tube from limit stop to limit stop. The angle of travel shall not exceed 190 degrees from the zero position.

4.4.5.16 Muzzle cover test. With the muzzle cover installed, a pull scale attached to the center of the cover handle, and a pulling force exerted in a direction perpendicular to the plane of the cover, the axial force required to remove each muzzle cover shall be 30 plus or minus 10 pounds.

4.4.5.17 Breakaway plug puller mechanism. Verify that the lever of the breakaway plug puller mechanism in each barrel can be latched and unlatched by hand without excessive force.

4.4.5.18 Pressure switch operation

4.4.5.18.1 With the ready light switch actuator plate rotated to the ON position and utilizing suitable means to indicate electrical continuity, gradually apply pressure to the air flask at a rate not exceeding 250 psig per minute until the pressure switch closes. The pressure switch in each barrel shall close at 1275 plus or minus 75 psig and remain closed at pressures above this value as indicated by a continuity reading.

4.4.5.18.2 Gradually decrease the pressure until the pressure switch opens. The pressure switch in each barrel shall open at 1200 psig minimum and remain open at pressures below this value.

4.4.5.19 Ready light switch operation. With the pressure switch closed as in 4.4.5.18.1, place the ready light switch actuator plate in the OFF position. The ready light switch shall be open, with no continuity reading indicated. This test shall be performed on each barrel.

4.4.5.20 Dry firing tests. Each breech mechanism shall be charged gradually at a rate not exceeding 250 psig per minute to pressures in accordance with Schedule I. With the control valve operating lever in the READY position, the plug puller mechanism in the unlatched position, the breakaway plug disconnected from the plug puller mechanism and secured to the guide, and all firing interlocks satisfied, each barrel shall be fired a total of six times without a torpedo in place. Four of the firings shall be made by electrical means and two by mechanical means. The power supply (4.4.4.10) shall be varied as specified in Schedule I for the four electrical firings. The application of voltage shall be regulated by the test circuit of 4.4.4.17, which shall limit firing potential duration to 400 milliseconds maximum. After each firing the breech mechanism shall be rotated to ensure that all three breech mechanisms will function in any of the three barrels.

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SCHEDULE I

Dry Firing Tests

Firing Sequence	Flask Pressure	AC Input Voltage to Soledoid	Pressure Switch Condition	Firing Method
1	1200 psig			Manual
2	2000 psig			Manual
3	1350 psig	102.5 ±0.5	Closed	Electrical
4	2000 psig	102.5 ±0.5	Closed	Electrical
5	1350 psig	123.5 ±0.5	Closed	Electrical
6	2000 psig	123.5 ±0.5	Closed	Electrical

NOTE : Prior to conducting the above tests remove the clevis pin from the lever of the plug puller mechanism and disconnect the lanyard from the lever. Replace and secure the clevis pin through the keeper and lever. Secure the breakaway plug to the guide by looping the lanyard over the plug.

* 4.4.5.20.1 Electrical firing test. Each barrel electrically fired shall fire with only one 400-millisecond application of test potential. Two firings shall be made at 102.5 plus or minus 0.5 volts ac and two at 123.5 plus or minus 0.5 volts ac.

4.4.5.20.2 Manual firing. During the manual firing tests measure the force required to activate the solenoid valve with the push scale. The force required shall be 25 pounds maximum.

4.4.5.20.3 Breakaway plug mechanism. After each dry firing, the plug puller mechanism shall be in the latched position and shall be capable of being unlatched by hand.

4.4.5.20.4 Control valve operating lever. During all dry firing tests observe that there is no leakage or premature air escape while the control valve operating lever is in the STANDBY condition. While performing the tests of Schedule I, with the control valve operating lever in the STANDBY

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condition, attempt to manually and electrically fire the tube. The tube shall not fire and there shall be no loss of pressure in the breech mechanism.

4.4.5.20.5 Pressure switch operation. During the dry firing tests verify that the pressure switch condition is in accordance with Schedule I when the control valve operating lever is in the READY condition.

4.4.5.20.6 Breakaway plug mechanism. The breakaway plug mechanism shall permit firing when latched in the breechward position and shall remain in the latched position at the completion of the firing.

4.4.5.20.7 Observation for damage. At the completion of the dry firing tests there shall be no damage sustained by the torpedo tube assembly.

4.4.5.21 Continuity. Continuity shall be in accordance with the applicable schematic diagram.

4.4..5.22 Lanyard retainer (breechward access opening location). The lanyard retainer plunger shall be depressed by hand and a loop of 0.069 minimum diameter wire shall be assembled to the retainer.

4.4.5.22.1 The plungers shall depress easily by hand pressure.

4.4.5.22.2 The retainers shall accept and secure the loop of 0.069 minimum diameter wire.

X 4.4.5.23 Torpedo exit velocity

The two photoelectric control units (4.4.4.21) shall be spaced three feet apart alongside the track or ramp. The first set of cells shall be located the length of the torpedo plus or minus 0.5 foot from the muzzle end of the barrel. The photoelectric cells shall be attached to the electronic counter (4.4.4.22) so that when the first beam is broken by the nose of the torpedo the counter starts and when the second beam is broken by the nose of the torpedo, the counter stops. The average velocity shall then be calculated by the formula $V = d/t$ where d = distance in feet, V = average velocity, and t = time in seconds. Each breech mechanism shall be charged at a rate not exceeding 250 psig per minute. The breech mechanism shall be charged to a pressure of 200 psi greater than those pressures specified in Schedule 11 and allowed to cool for at least 5 minutes prior to being bled and readied for a firing in accordance with Schedule II. The minimum exit velocity shall be met in all firings.

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SCHEDULE II

Torpedo Exit Velocity Tests

Launchings per Barrel (No.)	Torpedo Type (Mk)	Torpedo Type (Mod)	Air Flask Pressure (psig)	Pressure Switch Condition	Firing Method	Minimum Exit Velocity (feet/second)
1	44	1	1200		Manual	40
1	44	1	1350	Closed	Electrical	40
1	44	1	1500	Closed	Electrical	40
1	46	1	1200	-	Manual	40
1	46	1	1350	Closed	Electrical	40
1	46	1	1500	Closed	Electrical	40

* 4.4.5.23.1 Electrical firing test. Each electrical firing test shall be made with only one 400-millisecond application of test potential. The test potential used for these tests shall be 115 plus or minus 0.5 volts ac.

4.4.5.23.2 Manual firing. During the manual firing tests measure with the push scale the force required to actuate the solenoid valve. The force required shall be 25 pounds maximum.

4.4.5.23.3 Breakaway plug mechanism

* 4.4.5.23.3.1 Prior to each firing the breakaway plug mechanism lever shall be placed in the unlatched position and the breakaway plug attached to the torpedo receptacle.

* 4.4.4.23.3.2 After each firing the plug puller mechanism lever shall be in the latched postion and shall be capable of being unlatched by hand.

* 4.4.5.23.3.3 After each firing the breakaway plug shall be inspected for damage.

4.4.5.23.4 Pressure switch operation. bring the firing tests verify that continuity of the pressure switch circuit exists in accordance with Schedule II when the ready light switch is in the ON position and the control valve operating lever is in the READY condition.

4.4.5.23.5 Control valve operating lever. During all tests observe that when the control valve operating lever is in the STANDBY condition there is no leakage or premature air escape.

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4.4.5.23.6 Observation for damage. At the completion of the firing tests there shall be no damage to the barrel assemblies.

4.4.5.24 Packaging, packing, and marking. The packaging, packing, and marking of the torpedo tube assembly shall be examined to ensure that it complies fully with the requirements of Section 5.

4.5 Preproduction and periodic production tests. Each preproduction and periodic production sample, after satisfactorily passing the quality conformance tests and examinations (4.4), shall be subjected to the following environmental tests. Unless otherwise specified, the sample shall be deemed defective if it fails to satisfactorily pass the quality conformance tests and examinations subsequent to these environmental tests.

4.5.1 Vibration tests.

4.5.1.1 Vibration, nonoperation. The torpedo tube, secured to the mounting bracket of the testing machine in the same manner that it will be secured on shipboard, shall be subjected to the exploratory vibration test, variable frequency test, and endurance test as specified in MIL-STD-167 for Type I vibrations.

4.5.l.2 Vibration, standby condition. A dummy Mk 44 Mod 1 Torpedo shall be loaded in either of the two lower barrels of the torpedo tube and dummy Mk 46 Mod 1 Torpedoes shall be loaded in the remaining barrels. All three breech assemblies shall be pressurized to 1500 psig. The torpedo tube, loaded as specified and secured to the mounting bracket of the testing machine in the same manner that it will be secured on shipboard, shall be subjected to the exploratory vibration test, variable frequency test, and endurance test as specified in MIL-STD-167 for Type I vibrations.

4.5.2 Shock tests

* 4.5.2.1 Shock, nonoperating, The torpedo tube assembly shall be attached to the anvil table of the shock testing machine by means of a standard mounting platform and a 30 degree mounting fixture as specified in MIL-S-901 for medium-weight conditon. The torpedo tube shall be oriented on the 30 degree mounting fixture such that the barrel Centerlines are parallel to the horizontal plane. In each mounting condition the torpedo tube shall be subjected to Type A shock tests as specified in MIL-S-901 for medium-weight equipment.

4.5.2.2 Shock, loaded condition. The torpedo tube assembly shall be attached to the anvil table of the shock testing machine by means of a standard mounting platform and a 30 degree mounting fixture as specified in MIL-S-901 for medium-weight equipment. A dummy Mk 44 Mod 1 Torpedo shall be loaded in either of the two lower barrels of the torpedo tube and dummy Mk 46 Mod 1 Torpedoes shall be loaded in the remaining barrels. The Torpedo tube shall be oriented

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on the 30 degree mounting fixture such that the barrel centerlines are 30 degrees from the horizonal plane. In each mounting condition the torpedo tube shall be subjected to Type A shock tests as specified in MIL-S-901 for medium-weight equipment.

4.5.3 Temperature and humidity tests

* 4.5.3.1 Low temperature tests, loaded condition. Three dummy torpedoes shall be loaded in the torpedo tube. At least one dummy torpedo shall be of the Mk 44 Mod 1 type and at least one other dummy torpedo shall be of the Mk 46 Mod 1 type. Thermocouples shall be installed on the surface of the torpedoes in the area of contact between the torpedo and the barrel quadrants. Provisions shall be made for recording the sensed temperatures. The torpedo tube shall then be placed in a test chamber and 115 plus or minus 0.5 volts ac applied to terminals 1 and 2 of TB102. Ihe 125 volt dc power supply of 4.4.4.25 shall be applied to terminals 5 and 11 of TB2 (Mod 5) or terminals 10 and 13 of TB102 (Mod 7) and all heater switches shall be placed in the ON position. The temperature of the test chamber shall be lowered to -29 ± 3°C (-20 ± 5°F) and maintained at this temperature for a period of 48 plus or minus 5 hours. At the end of 24 hours in the test chamber, the base of the torpedo tube and the training gear joint shall be liberally sprayed with fresh water whose temperature is 0°C (33°F) At the end of 48 plus or minus 5 hours, the training gear shall be checked for operability, the temperatures shall be recorded for the three torpedoes, and the complete torpedo tube shall be performance checked. The temperatures recorded for the torpedoes shall at no time exceed 33.5°C (100°F) or be less than 0.5°c (33°F).

* 4.5.3.2 High temperature and humidity tests. The torpedo tube assembly, with fully pressurized breech assemblies (2000 psig), shall be placed within a test chamber and subjected to temperature and humidity cycling as specified in MIL-STD-81O, Method 507, Procedure IV with the following exceptions:

(a) Referenced measurements of steps 4, 5, 6, and 8 may be deleted.

(b) The 24-hour cycles of step 5 shall have a high temperature minimum of 59°C (138°F) and a maximum as specified.

(c) The torpedo tube assembly shall be performance checked during the high temperature portion of the fifth cycle of step 5 prior to decreasing to 30°c (86°F).

(d) Energize the barrel heater relay using the power supply of 4.4.4.25 applied to terminals 5 and 11 of TB2 (Mod 5) or terminals 10 and 13 of TB102 (Mod 7) and monitor the continuity between terminal 12 of TB102 and terminal 2 of TB103 during all cycles of step 5. Continuity shall exist between these terminals during the low temperature phase of the cycles and no continuity shall exist during the high temperature phase of the cycles.

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4.5.4 Marine environment

4.5.4.1 Marine atmosphere environment test. The torpedo tube assembly shall be exposed at dockside or on the weather deck of a ship for 60 plus or minus 5 days. The access covers and muzzle covers shall be removed for a period of 10 plus or minus 1 days. The assembly shall be protected from rain or snow by an appropriate shelter that will permit free circulation of air. At the completion of this test, there shall be no evidence of corrosion.

* 4.5.4.2 Alternative test for marine atmosphere. Subject the torpedo tube assembly to the Salt Fog test of MIL-STD-810, Method 509, using a 20 percent salt solution. The solution shall be prepared by dissolving 20 plus or minus 2 parts by weight of salt in 80 parts by weight of distilled water. The solution shall be adjusted to and maintained at a specific gravity between 1.126 and 1.157. At the completion of this test there shall be no evidence of corrosion.

4.5.5 During and after performance of the above environmental tests, no damage shall occur which could affect proper operation or function of the tube.

4.5.6 At the completion of the above environmental tests, the torpedo tube shall meet all the quality conformance tests of 4.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning. Unless otherwise specified, the torpedo tube, in the nonoperating condition, shall be cleaned in accordance with Method C-1 of MIL-P-116.

5.1.1.2 Drying. Unless otherwise specified, the torpedo tube shall be dried in accordance with one or more of the procedures of MIL-P-116, as appropriate.

5.1.1.3 Preservation. Immediately after cleaning and drying the torpedo tube, the external unpainted metal surfaces which are subject to corrosion shall be preserved by the application of preservative compound conforming to the requirements of MIL-C-16173, Grade 2.

5.1.1.4 Unit packaging. Not applicable.

5.1.2 Level B. Not applicable.

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5.1.3 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the supplier's commercial practice when such practices have been approved by the procuring activity.

5.2 Packing

5.2.1 Level A.

5.2.1.1 Exterior containers. The torpedo tube assembly, in the nonoperating condition, sha1l be packed in a nailed wood, skid - mounted crate with rubbing strips irJ accordance with MIL-C for Class I, Style A, Type I crates. The base of the torpedo tube shall be rigidly fastened to the load-bearing members of the skid base and the tube barrels shall be braced and cushioned to prevent movement.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The torpedo tube assembly, in the nonoperating condition, shall be packed to afford protection against damage during direct shipment from the supply soure to the first receiving activity for immediate use. Shipping containers shall comply with the carrier rules and regulations applicable to mode of transportation.

5.3 Marking

5.3.1 Special markings. None, unless otherwise specified (See 6.2).

5.3.2 Normal markings. In addition to the markings required by contract or order, shipping containers shall be marked in accordance with MIL-STD-129, including serial numbers.

* 5.4 Inspection data sheets. When Quality Conformance Inspection Data sheets are specified in the contract or order (see 6.2.2) for completion in accordance with 4.4.2, one data sheet (or copy thereof), bearing recorded data, shall be shipped with each torpedo tube assembly inspected. The torpedo tube assembly equpment serial number shall correspond to the serial number recorded on the accompanying data sheet

6. NOTES

6.1 Intended use. Torpedo Tube Assemblies Mk 32 Mod 5 and Mk 32 Mod 7 are used on the weather decks of Naval surface vessels to launch Torpedoes Mk 44 and Mk 46. Torpedo Tube Assembly Mk 32 Mod 5 is intended for use with Control Panel Mk 264 Mod 1 of fire Control Systems Mk 102, Mk 105 (except Mod 24), and Mk 111. Torpedo Tube Assembly Mk 32 Mod 7 is intended for use with Attack Console Mk 53 Mods 0 and of fire control System Mk 114.

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6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) The specific torpedo tube being procured (Mk 32 Mod 5 or Mk 32 Mod 7).

(c) When a preproduction sample is not required (See 3.2.1).

(d) When the Government waives the right to post-delivery inspection (See 4.2.4).

(e) Designated activity for preproduction and/or periodic production inspection(s) (See 4.3.1 and 4.3.2).

(f) Required level of preservation and packaging (See 5.1)

(g) Required level of packing (See 5.2).

(h) Whether anY special markings are required (See 5.3.1).

(i) Any exceptions to this specification or its applicable drawings and documents.

(j) That all additional samples required shall be provided and inspected at the expense of the supplier, when suck additional samples are required because Of failure of any sample(s) to pass the prescribed inspections.

6.2.2 Contract data requirements. Items 0f deliverable data required by this specification are cited in the following, paragraph herein:

Paragraph	Data Requirement	Applicable DIDS
4.4.2	Test Results	UDI-E-20602

DIDS (Data Item Description/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDS when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.2.3 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in MIL-P-116.

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Preparing Activity
NAVV - OS

Project No. 1045-NO07

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DATA SHEET

QUALITY CONFORMANCE INSPECTION

TESTS AND EXAMINATIONS

FOR

TORPEDO TUBE MARK 32 MODS 5 AND 7

(LD 162165 and LD 162166 respectively)

EQUIPMENT STATUS

CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser. No.		Title	
Insp. Req. Satisfied		Agency	

ORDALT STATUS

ORDALT NO.	INSTALLED	ORDALT NO.	INSTALLED

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DATA SHEET

QUALITY CONFORMANCE INSPECTION

TORPEDO TUBE MARK 32 MOD _____ Equipment Serial No. _____

10. SCOPE

10.1 This appendix contains data sheets to record the results of the Quality Conformance tests and examinations of 4.4 of this specification. When completed, with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the Torpedo Tube assembly bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test Sequence. When the contract or order specifies use of this appendix, the Quality Conformance Inspection tests of 4.4 shall be completed and the results recorded or verified in these date sheets prior to acceptance of the Torpedo Tube Assembly. unless otherwise noted, the sequence of testing procedures presented herein is suggested only. The procedures may be performed in any order which will permit the most practical and economical use of manpower and equipment.

20.2 Recording of results. Where space is provided in the “Reading” column of the Quality Conformance Test data sheets, the value or specific reading obtained shall be recorded and the initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated, the individual performing the test or inspection shall initial in the space provided in the “Reading” column indicating that the requirements have been met satisfactorily.

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	Reference	Limits	Reading
30. QUALITY CONFORMANCE INSPECTION TEST AND EXAMINATION DATA			
30.1 Test equipment	4.4.4	In calibration	_____
30.2 Test procedure	4.4.5		
30.2.1 4500 psig hydrostatic pressure, breech mechanism	4.4.5.1		
Barrel No. 1		Verified	_____
Barrel No. 2		Verified	_____
Barrel No. 3		Verified	_____
30.2.2 Pneumatic system pressure drop	4.4.5.2		
Securing mechanism removed			
Barrel No. 1		1350 psig min	_____
Barrel No. 2		1350 psig min	_____
Barrel No. 3		1350 psig min	_____
Securing mechanism installed			
Barrel No. 1		750 psig min	_____
Barrel No. 2		750 psig min	_____
Barrel No. 3		750 psig min	_____
30.2.3 Breech mechanism air loss	4.4.5.3		
Barrel No. 1		80.8 lbs max pressure loss	_____
Barrel No. 2		80.8 lbs max pressure loss	_____
Barrel No. 3		80.8 lbs max pressure loss	_____
30.2.4 Bore size	4.4.5.4		
Barrel No. 1		150 lbs max	_____
Barrel No. 2		150 lbs max	_____
Barrel No. 3		150 lbs max	_____
30,2.5 Dielectric strength	4.4.5.5.	No breakdown	_____

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	Reference	Limits	Reading
* 30.2.6 Insulation resistance	4.4.5.6		
Barrel Circuit		3 megohms min	_____
Training Gear Circuit		0.5 megohms min	_____
30.2.7 Ground circuit continuity	4.4.5.7	Verified continuity	_____
* 30,2.8 Training gear heater. resistance	4.4.5.8		_____
Resistance test	4.4.5.8.1	6.9+ 0.7 or -0.4 ohms	_____
Alternative current drain test	4.4.5.8.2	16.7 + 1.1 or -1.5 amps	_____
* 30.2.9 Barrel heater resistance	4.4.5.9		
Barrel No. 1		11.4 = 0.8 ohms	_____
Barrel No. 2		11.4 = 0.8 ohms	_____
Barrel No. 3		11.4 = 0.8 ohms	_____
* 30.2.10 Heater circuits operating current	4.4.5.10	50.5 amps max	_____
30.2.11 Squib fire switch	4.4.5.11		
Barrel No. 1		Proper operation	_____
Barrel No. 2		Proper operation	_____
Barrel No. 3		Proper operation	_____
30.2.12 Interlock switch	4.4.5.12	open 0° to 35° ± 2.5°	_____
		Open 145° ± 2.5° to 185″	_____
		Closed 35°± 2.5° tO 145°± 2.5°	_____
30.2.13 Training crank	4.4.5.13	30 lbs maX	_____
30.2.14 Retractable stop	4.4.5.14	45°± 2.5°	_____
30.2.15 limit stops	4.4.5.15	190° max	_____
30.2.16 Muzzle cover	4.4.5.16		
Barrel No. 1		30 ± 10 lbs	_____
Barrel No. 2		30 ± 10 lbs	_____
Barrel No. 3		30 ± 10 lbs	_____

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	Reference	Limits	Reading
30.2.17 Breakaway plug puller mechanism	4.4.5.17		
Barrel No. 1		Hand Latchable, Hand Unmatchable	_____
Barrel No. 2		Hand Latchable, Hand Unmatchable	_____
Barrel No. 3		Hand Latchable, Hand Unmatchable	_____
30.2.18 Pressure switch	4.4.5.18		
Barrel No. 1 switch closure		1275 ± 75 psig	_____
Barrel No. 1 switch opening		1200 psig min	_____
Barrel No. 2 switch closure		1275 ± 75 psig	_____
Barrel No. 2 switch opening		1200 psig min	_____
Barrel No. 3 switch closure		1275 ± 75 psig	_____
Barrel No. 3 switch opening		1200 psig min	_____
30.2.19 Ready light switch	4.4.5.19		
Barrel No. 1		Switch opens	_____
Barrel No. 2		Switch opens	_____
Barrel No. 3		Switch opens	_____
30.2.20 Dry firing test	4.4.5.20	Breech mechanisms function in all barrels	_____
Electrical firing	4.4.5.20.1		
Barrel No. 1		One 400-Millisecond voltage application per firing	_____
Barrel No, 2		One 400-Millisecond voltage application per firing	_____
Barrel No. 3		One 400-Millisecond voltage application per firing	_____

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	Reference	Limits	Reading
Manual firing	4.4.5.20.2		
Barrel No. 1		25 lbs max	
Barrel No. 2		25 lbs maX	
Barrel No. 3		25 lbs maX	
Breakaway plug mechanism	4.4.5.20.3		
Barrel No. 1		Latched, hand Unmatchable	
Barrel No. 2		Latched, hand Unmatchable	
Barrel No. 3		Latched, hand Unmatchable	
Cover valve operating lever	4.4.5.20.4		
Barrel No. 1		No leakage, will not fire	
Barrel No. 2		No leakage, will not fire	
Barrel No. 3		No leakage, will not fire	
Pressure switch operation	4.4.5.20.5		
Barrel No. 1		Operation per Schedule I	
Barrel No. 2		Operation per Schedule I	
Barrel No. 3		Operation per Schedule I	
Breakaway plug mechanism	4.4.5.20.6		
Barrel No. 1		Permits firing, remains latched	
Barrel No. 2		Permits firing, remains latched	
Barrel No. 3		Permits firing, remains latched	

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	Reference	Limits	Reading
Observation for damage	4.4.5.20.7		
Barrel No. 1		No damage	
Barrel No. 2		No Damage	
Barrel No. 3		No Damage	
30.2.21 Continuity	4.4.5.21	Continuity per schematic	
30.2.22 Lanyard retainers	4.4.5.22		
Barrel No. 1		Hand depressable, secures wire loop	
Barrel No. 2		Hand depressable, secures wire loop	
Barrel No. 3		Hand depressable, secures wire loop	

30.2.23 Torpedo exit velocity (3 barrels) 4.4.5.23

Barrel No. 1

Launchings per barrel (No.)	Torpedo Type (Mk)	(Mod)	Air Flask Pressure (psig)	Pressure Switch Condition	Firing Method	Min. Exit Velocity (Ft/sec)	Reading Obtained
1	44	1	1200		Manual	40	
1	44	1	1350	Closed	Electrical	40	
1	44	1	1500	Closed	Electrical	40	
1	46	1	1200	-	Manual	40	
1	46	1	1350	Closed	Electrical	40	
1	46	1	1500	Closed	Electrical	40	

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	Reference
Barrel No. 2	4.4.5.23

Launchings per Barrel (No.)	Torpedo Type (Mk)	(Mod)	Air Flask Pressure (psig)	Pressure, Switch Condition	Firing Method	Min. Exit Velocity (Ft/see)	Reading Obtained
1	44	1	1200		Manua1	40	
1	44	1	1350	Closed	Electrical	40	
1	44	1	1500	Closed	Electrical	40	
1	46	1	1200		Manual	40	
1	46	1	1350	Closed	Electrical	40	
1	46	1	1500	Closed	Electrical	40	

	Reference
Barrel No. 3	4.4.5.23

Launchings per barrel (No.)	Torpedo Type (Mk)	(Mod)	Air Flask Pressure (psig)	Pressure Switch Condition	Firing Method	Min. Exit Velocity (Ft/sec)	Reading Obtained
1	44	1	1200	-	Manual	40	
1	44	1	1350	Closed	Electrical	40	
1	44	1	1500	Closed	Electrical	40	
1	46	1	1200	-	Manual	40	
1	46	1	1350	Closed	Electrical	40	
1	46	1	1500	Closed	Electrical	40	

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	Reference	Limits	Reading
Electrical firing	4.4.5.23.1		
Barrel No. 1		One 400-millisecond voltage application per firing	_____
Barrel No. 2		One 400-millisecond voltage application per firing	_____
Barrel No. 3		One 400-millisecond voltage application per firing	_____
Manual firing	4.4.5.23.2		
Barrel No. 1		25 lbs maX	_____
Barrel No. 2		25 lbs max	_____
Barrel No. 3		25 lbs maX	_____
Breakaway plug mechanism	4.4.5.23.3		
Breakaway plug mechanism	4.4.5.23.3.1		
Barrel No. 1		Unlatched, attached to torpedo recep-tacle	_____
Barrel No. 2		Unlatched, attached to torpedo recep-tacle	_____
Barrel No. 3		Unlatched, attached to torpedo recep-tacle	_____
Breakaway plug	4.4.5.23.3.2		
Barrel No. 1		Latched, hand Unmatchable	_____
Barrel No. 2		Latched, hand Unmatchable	_____
Barrel No. 3		Latched, hand Unmatchable	_____

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	Reference	Limits	Reading
Breakaway plug	4.4.5.23.3.3		
Barrel No. 1		No damage after each firing	_____
Barrel No. 2		No damage after each firing	_____
Barrel No. 3		No damage after each firing	_____
Pressure switch operation	4.4.5.23.4		
Barrel No, 1		Operation per Schedule II	_____
Barrel No. 2		Operation per Schedule II	_____
Barrel No. 3		Operation per Schedule II	_____
Control valve operating lever	4.4.5.23.5		
Barrel No. 1		No leakage	_____
Barrel No. 2		No leakage	_____
Barrel No. 3		No leakage	_____
Observation for damage	4.4.5.23.6		
Barrel No. 1		No damage	_____
Barrel No. 2		No damage	_____
Barrel No. 3		No damage	_____
30.3 Packaging, packing, and marking			
30.3.1 Preservation and packaging	5.1		_____
30.3.2 Packing	5.2		_____
30.3.3 Marking	5.3		_____

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